

Generated benchmark artifacts

Typeset from `paper/generated/aarch64/` by `benchmarks/scripts/build_pdf.py`. Every number resolves to a row in the result set named below; see `PROVENANCE.md` for the inputs, transformation and caveats of each artifact.

```
result set: current
commit: 00d8c08aa656
  host: galaxy
measured: 2026-08-09
inputs: sha256:884176aa0dedf313
```

Dataset	Tool	Metric	-@	Mapped	Mapq 60	Missed (%)	Wrong Q60	Index (s)	Map (s)	Mem (GB)
B01	shmap-rs	Containment	1	149 194	139 309	6.78	—	6.58	35.99	2.63
B01	shmap-rs	Jaccard	1	146 120	138 421	7.37	—	6.47	50.77	2.67
B01	shmap-rs	bucket_SH	1	149 236	138 250	7.49	—	6.50	27.43	2.65
B01	mapquik	—	32	48 913	—	—	—	—	19.50	19.37
B01	winnowmap2	—	32	189 022	—	—	—	—	1328.30	9.78
B02	shmap-rs	Containment	1	125 000	117 532	5.97	0	6.89	28.20	2.59
B02	shmap-rs	Jaccard	1	125 000	119 065	4.75	6	6.51	43.93	2.66
B02	shmap-rs	bucket_SH	1	125 000	116 800	6.56	1	6.54	24.21	2.66
B02	mapquik	—	32	39 965	—	—	—	—	19.20	19.38
B02	winnowmap2	—	32	129 488	—	—	—	—	1006.40	9.40
B03	shmap-rs	Containment	1	241 991	227 521	6.19	—	6.49	45.17	2.66
B03	shmap-rs	Jaccard	1	241 265	228 887	5.63	—	6.50	61.26	2.65
B03	shmap-rs	bucket_SH	1	242 050	226 931	6.43	—	6.48	36.00	2.65
B03	mapquik	—	32	89 559	—	—	—	—	20.90	19.42
B03	winnowmap2	—	32	280 093	—	—	—	—	997.00	8.04
B04	shmap-rs	Containment	1	2 419 796	2 273 122	6.28	—	6.49	456.58	2.65
B04	shmap-rs	Jaccard	1	2 411 461	2 286 085	5.74	—	6.47	623.40	2.65
B04	shmap-rs	bucket_SH	1	2 420 331	2 267 229	6.52	—	6.47	358.52	2.65
B04	mapquik	—	32	897 326	—	—	—	—	44.80	19.36
B04	winnowmap2	—	32	2 808 979	—	—	—	—	9019.50	8.38
B05	shmap-rs	Containment	1	39 608	37 839	58.97	—	6.54	13.74	2.59
B05	shmap-rs	Jaccard	1	5 956	5 409	94.13	—	6.49	16.94	2.65
B05	shmap-rs	bucket_SH	1	41 115	39 226	57.46	—	6.52	12.55	2.66
B05	winnowmap2	—	32	99 107	—	—	—	—	331.50	6.15

Table 1: Long-read mapping tools on the benchmark suite. Mapq 60 is the count of confident mappings; Missed is the share of input reads either unmapped or below mapq 60; Wrong Q60 counts confident mappings that are wrong against ground truth.

Dataset	Metric	Matches per read			Potential		Buckets per read	
		possible	examined	in mapping	realized	unrealized	seeded	final

Table 2: Efficiency of the seed heuristic. Realized potential is possible matches divided by matches examined (work avoided); unrealized potential is matches examined divided by matches inside the reported mapping (work still wasted).

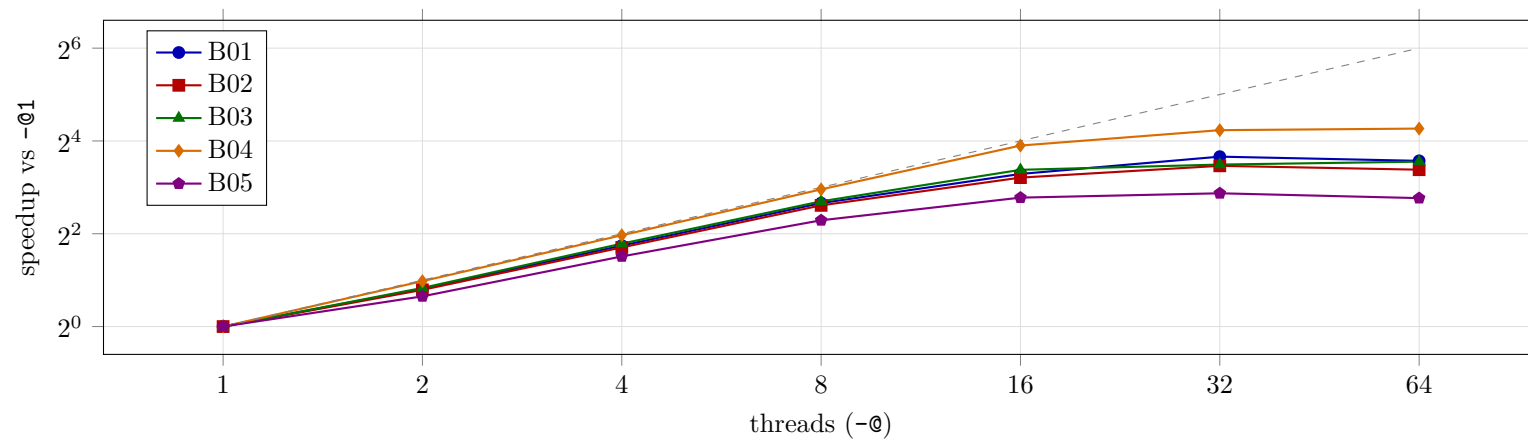


Figure 1: Whole-run speedup against thread count, Containment. The dashed line is linear speedup.

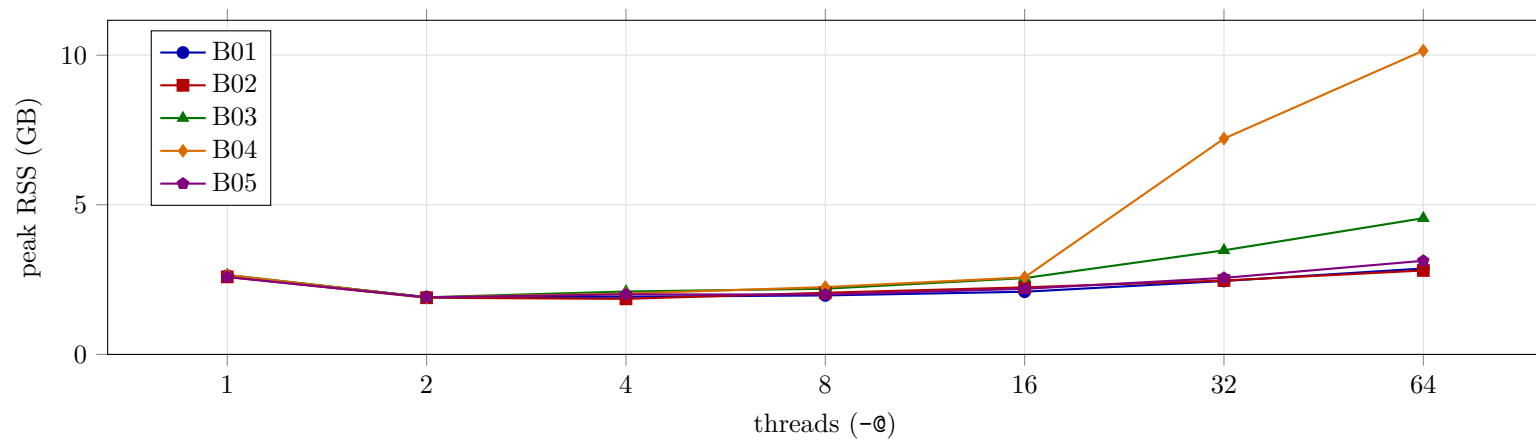


Figure 2: Peak resident memory against thread count, Containment, with the C++ reference as a horizontal line.

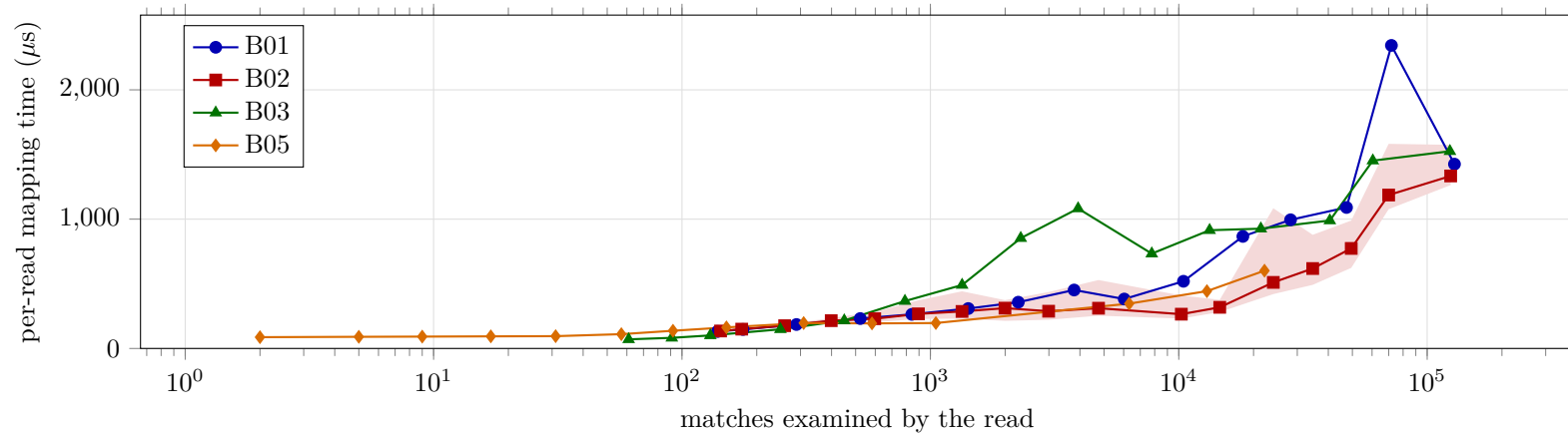


Figure 3: Per-read mapping time against the number of matches the read examined, Containment. Points are per-bin medians over reads grouped into 18 logarithmic bins; the shaded band is the interquartile range for the simulated set.

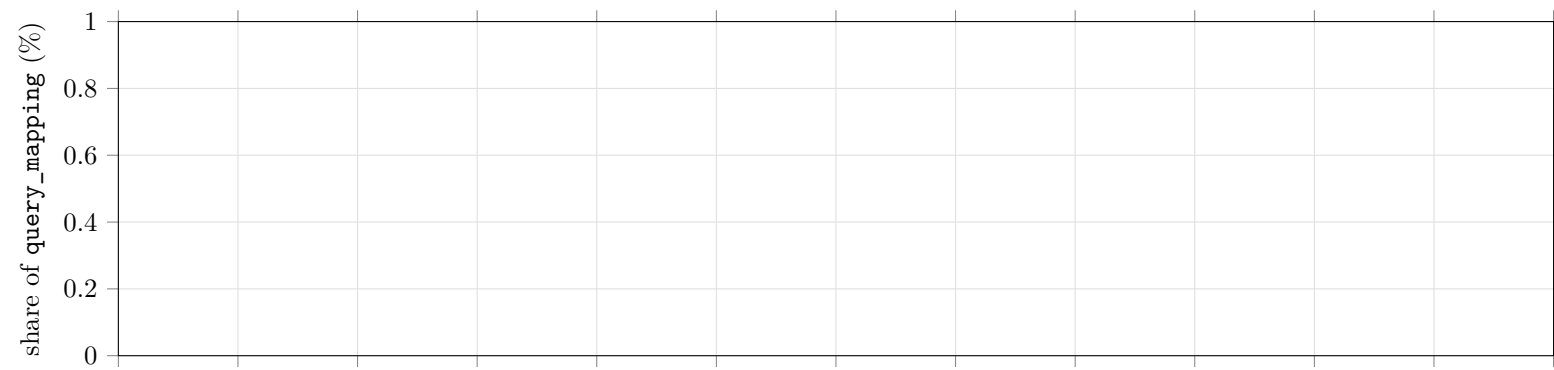


Figure 4: Where mapping time goes: stage shares of `query_mapping`, single-threaded.